

课堂作业整理 3.24

例 5-1 (理想)

$$\frac{V_s}{10k} = \frac{2}{10k} = 0.2mA \quad - \quad \frac{V_s}{10k} = \frac{V_o}{20k} \quad - \quad \frac{V_o}{V_s} = 2$$

练 5-1

$$V_s + \frac{V_s}{5} \times 40 = V_o, \quad \text{即} \quad \frac{V_o}{V_s} = 9, \quad i_o = \frac{1}{5} + \frac{9}{20} = 0.65mA$$

例 5-4

$$\frac{V_a - V_o}{40k\Omega} = \frac{6 - V_a}{20k\Omega}, \quad V_a - V_o = 12 - 2V_a \Rightarrow V_o = 3V_a - 12$$

$$\text{又由 } V_a = V_o = 2V, \quad V_o = 6 - 12 = -6V$$

练 5-4

$$\text{在节点 2: } i_s = \frac{0 - V_1}{R_1}, \quad \text{即 } V_1 = -i_s R_1$$

$$\text{在节点 1: } \frac{0 - V_1}{R_1} = \frac{V_1 - 0}{R_2} + \frac{V_1 - V_o}{R_3} \Rightarrow -V_1 \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) = \frac{-V_o}{R_3}$$

$$\Rightarrow V_o = i_s R_1 R_3 \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) \Rightarrow \frac{V_o}{i_s} = -R_1 \left(1 + \frac{R_3}{R_1} + \frac{R_3}{R_2} \right)$$

例 5-6

$$\text{由 } V_o = - \left(\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right), \quad \text{则 } V_o = - \left[\frac{10}{5} \times 2 + \frac{10}{2.5} \times 1 \right] = -8V$$

$$\text{则 } i_o = \frac{V_o - 0}{10} + \frac{V_o - 0}{2} = -0.8 - 4 = -4.8mA$$

练 5-6

$$\frac{1.5 - 0}{20k} + \frac{2 - 0}{10k} + \frac{1.2 - 0}{6k} + \frac{V_o - 0}{8k} = 0$$

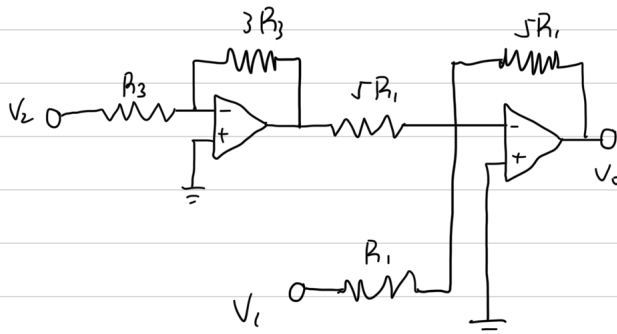
$$\text{得 } V_o = -3.8V$$

$$i_{8k} = \frac{V_o - 0}{8k} = \frac{-3.8}{8k} = -0.475mA$$

$$i_{4k} = \frac{V_o - 0}{4k} = \frac{-3.8}{4k} = -0.95mA$$

$$\text{则 } i_o = -0.475 - 0.95 = -1.425mA$$

例 5-7



复习题

1. c

2. c, d

3. b

4. b